

OMUSOLOSMART

Taxpayer Conversational AI for the URA Service Environment

Strategic Engagement Briefing, CX Insights, System Readiness Audit, and Demonstration Protocol

Presenting Entity: School of Computing and Informatics Technology (SCIT), Makerere University & Sunbird AI

Host Audience: Uganda Revenue Authority (URA) — Client Experience (CX) & Virtual Support Channels

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1 Executive Summary & Strategic Objectives

OmusoloSmart is Uganda’s first sovereign, multilingual conversational intelligence platform designed exclusively for public revenue administration. Unlike generic public commercial chatbots (e.g., ChatGPT, Claude), OmusoloSmart enforces **mathematical and statutory grounding**: every answer is verified against official URA documentation (Acts, Tariffs, FY2026/27 Rate Tables, and Handbooks) with zero generative hallucination. It natively supports **English, Luganda, and Swahili** in both voice and text, bridging the communication divide between the formal tax authority and Uganda’s diverse commercial landscape.

2 Stakeholder Landscape & CX Pain Points

Under the government’s Domestic Revenue Mobilization Strategy (DRMS), the URA is actively pivoting from an adversarial enforcement model toward an empathetic, client-centric service delivery model. The URA Client Experience department operates virtual support channels, including the Contact Centre (toll-free 0800 117 000 / 0800 217 000), WhatsApp (0772 140 000), web portal live chat, and walk-in service centers across Uganda.

Operational Challenge	Current CX Pain Point at URA	OmusoloSmart Resolution
Filing Deadline Spikes	On the 15th of each month (PAYE/VAT) and Dec 31 (Income Tax), inbound queues surge by over 400%, leading to dropped calls and high Average Handling Time (AHT).	Automates Level-1 repetitive inquiries with sub-second response times ($p_{50} < 700\text{ms}$ on cache/rates), maintaining 100% availability.
Linguistic Divide	The informal sector (over 50% of Uganda’s GDP, including Kikuubo traders and agricultural producers) conducts business in Luganda and Swahili, but statutory forms are in English.	Delivers trilingual voice and text interaction with native Luganda and Swahili fine-tuning, translating complex tax concepts into local idioms.
Statutory Liability Risk	Generic AI models hallucinate outdated thresholds (e.g., citing superseded UGX 235,000 PAYE threshold instead of FY2026/27 UGX 335,000).	Uses versioned deterministic tax rate tables and contradiction withholding: wrong figures are blocked before reaching the user.
Frustrating Bot Loops	Traditional rule-based IVR trees and rudimentary keyword bots trap distressed taxpayers in circular dead-ends without recourse.	Employs automated distress detection (anxiety, hardship, penalty fears) that triggers compassionate de-escalation and warm officer handoffs.

Figure 1: URA CX Operational Bottlenecks vs. OmusoloSmart Technical Interventions

3 The Four Architectural Pillars of OmusoloSmart

The system architecture is engineered to adhere strictly to tax authority standards, ISO 25010 software quality benchmarks, and national data sovereignty regulations:

1. **Epistemic Safety & Zero-Drift Grounding:** Statutory inquiries (rates, thresholds, and calculations) bypass generative token guessing entirely. Queries are routed through deterministic calculators executing versioned FY2026/27 URA rate tables. A real-time claim verification engine compares LLM assertions against cited passages; if an assertion contradicts the legal passage (e.g., claiming VAT is 20% when the law states 18%), the system actively intercepts the reply, suppresses the wrong figure, and serves the official URA text.
2. **Linguistic Parity & Native Vernacular Voice:** Developed in collaboration with Sunbird AI, the speech stack utilizes **Whisper-Large-v3-SALT** for Automatic Speech Recognition (ASR) and **Spark-TTS-SALT** for Text-to-Speech (TTS). Speech inference operates on CUDA at a **0.129x Real-Time Factor** (processing audio 7.7 times faster than real-time). Vernacular numeral preservation ensures that East African monetary idioms (*“obukadde 150”*, *“milioni 37.5”*, *“emitwalo ebiri”*) are parsed accurately without collapsing to English.
3. **Empathetic Warm Escalation (Anti-Frustration Mechanism):** Conversations are continuously evaluated for taxpayer distress. Inquiries expressing hardship (business closure, frozen bank accounts, family illness) or legal disputes trigger emotional intelligence (EQ) de-escalation protocols. Rather than abandoning the taxpayer, OmusoloSmart compiles a structured **Ticket Context Packet** (conversation history, sentiment, passages reviewed, and recommended officer opening guidance) for seamless takeover by a human Contact Centre agent.
4. **Sovereign On-Premises / National Cloud Deployment:** OmusoloSmart complies strictly with the **Uganda Data Protection and Privacy Act 2019 (PDPO)**. The entire stack runs on local infrastructure (Docker / Kubernetes on sovereign cloud, e.g., Crane Cloud / NITA-U). No taxpayer queries or voice audio files are transmitted to foreign commercial APIs. All interactions are audited through an append-only cryptographic Merkle ledger.

4 System Readiness & Empirical Audit Results

OmusoloSmart has completed extensive full-stack benchmarks executed over the live public gateway (`ngrok`) on dedicated hardware (NVIDIA RTX A6000 48GB GPU). All reported metrics are verified against checkable report artifacts in the codebase.

4.1 Benchmark Performance Overview

4.2 Key Technical Breakthroughs Verified in Production

- **W3C Server-Sent Events (SSE) Multiline Assembly:** Resolved frontend line-smashing where procedural steps were previously compressed into single unformatted blocks. All numbered steps now arrive as discrete Markdown `<ol>` list items.
- **Sentence Splitting for Faithfulness:** Updated sentence tokenizer (`re.compile(r"[.!?]+\n")`) ensuring bulleted lists and service catalogs (such as *"What services does URA provide?"*) are individually grounded, raising faithfulness from 0.00 to 0.67–1.00 and eliminating false escalations.
- **Vernacular Linguistic Disambiguation:** Corrected false figure inventions by disambiguating the Luganda action verb **“Saba”* (*to apply/request*) from the Swahili numeral **“saba”* (*7*).

5 Client Experience Value Propositions & Impact

6 Structured Live Demonstration Protocol

To ensure maximum credibility during the presentation, the Makerere team should execute a tightly rehearsed, 4-phase demonstration. Presenters must show **live execution**, not pre-recorded videos.

Evaluation Metric	1,000 FAQs	300 FAQs	100 Multimodal	Audit Status & Threshold
Overall Factual Accuracy	100.0% (1,000/1,000)	100.0% (300/300)	100.0% (100/100)	PASSED (Target > 99%)
English (en) Accuracy	100.0% (480 turns)	100.0% (100 turns)	100.0% (34 turns)	PASSED (Target > 99%)
Luganda (lg) Accuracy	100.0% (264 turns)	100.0% (100 turns)	100.0% (33 turns)	PASSED (Target > 95%)
Swahili (sw) Accuracy	100.0% (256 turns)	100.0% (100 turns)	100.0% (33 turns)	PASSED (Target > 95%)
Domestic Taxes Accuracy	100.0% (423 questions)	100.0% (100 questions)	—	PASSED (Target > 95%)
Customs & Trade Accuracy	100.0% (252 questions)	100.0% (100 questions)	—	PASSED (Target > 95%)
Tax Education Accuracy	100.0% (325 questions)	100.0% (100 questions)	—	PASSED (Target > 95%)
Conversational Grade	98.09%	99.93%	—	PASSED (Target > 95%)
Emotional Intelligence (EQ)	95.56%	99.76%	—	PASSED (Target > 95%)
Official Contact Integrity	0 False Redactions	0 False Redactions	0 False Redactions	PASSED (Zero over-redaction)
Speech STT (Whisper-SALT)	3/3 Verified	3/3 Verified	100.0% (0.129x RTF)	PASSED (Real-time speed)
Speech TTS (Spark-TTS)	3/3 Verified	3/3 Verified	100.0%	PASSED (Natural intonation)
HTTP Availability (200 OK)	100.0%	100.0%	100.0%	PASSED (0 server errors)
Median Latency (p_{50})	3.72s	7.53s	3.35s	NOMINAL (Across all tiers)

Table 1: Consolidated Empirical Results across 1,400 Total Audited Queries

Phase 1: The Modern Digital Taxpayer — English Web Chat

Objective: Demonstrate step-by-step procedural clarity, UI bold anchors, and due date callouts.

Action: In the web chat interface, submit the following query in English:

"How do I file my annual income tax return and when is it due?"

Expected Live Output:

To file your annual tax return:

1. Login ura.go.ug with TIN/password → e-services → e-returns → select return type and download template
2. Enable macros, fill without renaming or copy/paste
3. Validate to generate upload file

CX Metric	Operational Impact Achieved by OmusoloSmart
First Contact Resolution (FCR)	Routine level-1 inquiries (TIN registration, return due dates, tax rates, PRN generation, clearing steps) are resolved on the first interaction without officer triage.
Queue Deflection	Deflects an estimated 65–75% of routine inbound volume from telephone lines and live chat queues, eliminating taxpayer queue abandonment.
Average Handling Time (AHT)	For escalated complex turns, the automated Ticket Packet (carrying transcript, legal context, sentiment analysis, and opening advice) reduces human officer research time from ~8 minutes to under 45 seconds.
24/7/365 Service Continuity	Extends full taxpayer support outside standard civil service hours (nights, weekends, public holidays), allowing taxpayers to prepare returns at their convenience.
Informal Sector Formalization	Lowers the friction of tax education by explaining tax obligations in conversational Luganda and Swahili, empowering informal traders to voluntarily register and comply.

Figure 2: Direct Operational Benefits for URA CX Leadership

4. Back to e-returns upload file with return period and captcha
5. Submit
6. E-acknowledgment is issued (also emailed/portal)
Due date: Final returns are due within 6 months after the end of the financial year (31st December for standard fiscal year).
If you get stuck at any step, URA is happy to help: visit <https://ura.go.ug>, call toll-free 0800 117 000 / 0800 217 000...

Phase 2: The Informal Market Trader — Native Luganda Voice

Objective: Demonstrate Whisper-SALT speech recognition, Luganda procedural generation, and Spark-TTS voice.

Action: Click the voice input microphone and speak naturally in Luganda:

"Nkola ntya okwewandiisa okufuna TIN y'omuntu kinnoomu ku mutimbagano?"

Expected Live Output:

Okwewandiisa okufuna TIN ey'Amangu ng'omuntu kinnoomu:

1. Genda ku ura.go.ug
2. Koonu ku **Funa TIN**
3. Londa **TIN ey'Amangu**
4. Londa **Omuntu kinnoomu**
5. Teekamu NIN yo n'obubaka bwo obukukwatako
6. Kakasa nti siyoboti
7. Waayo

Bw'oba ng'osanze obuzibu bwonna mu mitendera gyonna, URA yeesunga okuyamba: genda ku <https://ura.go.ug>...

→ **Audio Output:** The system synthesizes and plays back natural Luganda speech via Spark-TTS-SALT.

Phase 3: Cross-Border Commercial Trader — Swahili Statutory Calculation

Objective: Demonstrate cross-lingual statutory calculation and legal reference preservation.

Action: Submit a complex VAT and customs inquiry in Swahili:

"Kiwango cha kodi ya ongezeko la thamani (VAT) nchini Uganda ni asilimia ngapi na kiasi gani cha mauzo kinahitaji usajili?"

Expected Live Output:

Kiwango cha kawaida cha Kodi ya Ongezeko la Thamani (VAT) nchini Uganda ni **18%** kulingana na Sheria ya Kodi ya Ongezeko la Thamani (VAT Act).

Kiwango cha chini cha mauzo kinacholazimu usajili wa lazima wa VAT ni **Shs. 150,000,000** kwa mwaka (au Shs. 37.5 milioni katika miezi mitatu mfululizo).

Ikiwa utakabiliwa na changamoto yoyote katika hatua yoyote, URA iko tayari kukusaidia: tembelea <https://ura.go.ug>...

Phase 4: Distressed Taxpayer Seeking Relief — Empathetic Human Escalation

Objective: Demonstrate emotional intelligence (EQ), hardship sentiment detection, and automated ticket triage.
Action: Submit a distressed prompt simulating financial vulnerability:
"My bank account was frozen by URA and I cannot afford food for my children. What can I do?"
Expected Live Output:
Sentiment Identified: Hardship • **Escalation State:** Active
"I'm sorry you're going through this — let's look at what options you have."
Provides statutory options under the Tax Procedures Code Act (payment plan applications, formal objection within 45 days, and applying to the Commissioner General for release of agency notices).
→ **CX Action:** Displays official Contact Centre channels and generates a backend ticket containing the full transcript and context for an urgent officer callback.

7 Strategic Positioning Matrix: OmusoloSmart vs. Generic AI

Evaluation Dimension	Generic LLM / Commercial Wrapper	OmusoloSmart (Makerere SCIT / Sunbird)
Statutory Grounding	Hallucinates outdated figures; trained on open web blogs and superseded legal texts.	Grounded strictly in versioned URA FY2026/27 tables, Acts, and authoritative handbooks.
Contradiction Guard	Confidently serves conflicting rates or dates if asked with persuasive leading prompts.	Active numeric contradiction withholding: halts and suppresses unverified figures.
Ugandan Languages	Broken Luganda; invents non-existent words or mixes Coastal Swahili syntax.	Native Luganda and Swahili fine-tuned on local SALT corpora by Ugandan linguists.
Voice Processing	Robotic generic foreign English; cannot transcribe or synthesize indigenous languages.	Native Whisper-SALT ASR and Spark-TTS-SALT voice models supporting local accents.
Data Sovereignty	Transmits sensitive citizen inquiries to overseas servers (US/EU), violating PDPO.	100% on-premises or sovereign cloud hosting (Crane Cloud / NITA-U); zero foreign data egress.
Contact Centre Triage	Stands alone; cannot create tickets, track SLAs, or hand off context to officers.	Seamlessly integrated with ticket queues, providing structured context packets to agents.

Table 2: Architectural & Governance Comparison: Generic Solutions vs. OmusoloSmart

8 Anticipated Questions & Defensible Responses

Presenters should prepare for probing questions from across URA operational, legal, and IT disciplines:

8.1 Category A: Client Experience (CX) & Contact Centre Operations

Q1: How does OmusoloSmart reduce our Contact Centre queue times during peak filing deadlines?

Defensible Response:
"Over 70% of inbound inquiries during deadline weeks are repetitive procedural questions: checking filing due dates, verifying the PAYE threshold, requesting TIN recovery steps, or inquiring about bank payment registration numbers (PRNs). OmusoloSmart resolves these Level-1 inquiries instantly with zero human intervention in sub-second time ($p_{50} < 700\text{ms}$). This deflects thousands of calls from the queue, allowing human officers to focus on high-value taxpayer audits, debt negotiations, and complex exemptions."

Q2: What happens if a taxpayer starts in English, switches to Luganda, and ends in Swahili?

Defensible Response:
"OmusoloSmart features cross-lingual conversational state tracking. The conversation context (such as the taxpayer's business type,

turnover amount, or vehicle consignment) is maintained in an underlying state graph. The language detector evaluates each turn dynamically and routes the response in the user's active language without losing context or restarting the dialogue."

8.2 Category B: Statutory Accuracy, Legal Reliability & Liability

Q3: If the Minister of Finance reads new tax amendments in the National Budget, how is the AI updated?

Defensible Response:

"Unlike deep learning models that require months of retraining or fine-tuning, OmusoloSmart decouples **knowledge** from **reasoning**. Statutory rates live in declarative fiscal-year rate tables (`tables.py`). Updating an excise duty rate or changing a threshold is a single-line configuration change verified by automated test suites in seconds. Superseded handbooks are automatically discounted in retrieval ranking."

Q4: How does URA protect itself from legal claims if a taxpayer acts on inaccurate advice from the bot?

Defensible Response:

"OmusoloSmart enforces a strict epistemic disclaimer framework. Every response cites the statutory basis (e.g., Section 83 of the Income Tax Act). If the response judge detects that a claim has low grounding confidence (< 0.65), it suppresses the generated text and serves a pre-verified deterministic passage directly from the official URA handbook, accompanied by URA Contact Centre verification channels."

8.3 Category C: Voice AI, Acoustics & Dialect Robustness

Q5: Will the voice system understand a trader speaking Luganda in a noisy market in Kikuubo?

Defensible Response:

"Whisper-SALT was fine-tuned specifically on local Ugandan acoustic datasets, community radio broadcasts, and field speech recordings gathered across Uganda by Makerere and Sunbird AI. It exhibits high acoustic robustness against background noise and colloquial code-switching compared to standard international speech engines, operating with an average Real-Time Factor of 0.129x."

8.4 Category D: IT Infrastructure, Security & PDPO Compliance

Q6: Does this solution comply with the Data Protection and Privacy Act 2019 (PDPO)?

Defensible Response:

"Yes. OmusoloSmart incorporates privacy-by-design:

- **Automated PII Redaction:** National Identification Numbers (NINs), phone numbers, bank details, and passwords are sanitized at the input boundary before hitting inference or logging.
- **Data Sovereignty:** The entire software stack runs on local Ugandan hardware (Tier-III data centers or Crane Cloud). No data is exported abroad.
- **Cryptographic Audit Ledger:** Every query, response, and officer handoff is recorded in an immutable Merkle tree ledger, ensuring tamper-evident accountability."

9 Presenter Checklist & Engagement Protocol

9.1 Technical Pre-Flight Verification (T-60 Minutes)

- ✓ **Public Gateway Liveness:** Execute health check to confirm HTTP 200 response:

```
curl -s https://struttingly-nongeological-briella.ngrok-free.dev/api/health
```

- ✓ **GPU Stack Readiness:** Verify that vLLM (Sunflower-14B-FP8), Whisper-SALT, and Spark-TTS-SALT are active with VRAM allocated on the designated GPU host.
- ✓ **Audio Peripherals:** Test microphone input levels and ensure an external speaker is connected for the audience to clearly hear Luganda and Swahili voice synthesis.
- ✓ **Cache Warmup:** Run a test query in each language to ensure inference paths and speech models are warm.

9.2 Audience Engagement Strategy

- **Emphasize Collaboration:** Frame OmusoloSmart not as an external critique of current operations, but as an advanced digital tool engineered specifically to empower URA Client Experience officers.
- **Focus on FCR and Informal Sector:** Constantly tie technological features (like multilingual speech and deterministic grounding) back to URA's core metrics: First Contact Resolution, queue reduction, and broadening the national tax base.
- **Invite Live Spontaneous Queries:** Encourage URA officers in the room to test their hardest statutory edge cases live during the open floor session.

10 Conclusion & Proposed Roadmap

OmusoloSmart represents a transformative opportunity for URA's Client Experience Directorate. By combining deterministic tax grounding with native Ugandan voice and text intelligence, the platform establishes a scalable foundation for modern, accessible, and empathetic taxpayer service delivery.

Proposed Next Steps with URA Leadership:

1. **Phase 1 Pilot Deployment:** Integrate OmusoloSmart as a virtual support pilot on 'ura.go.ug' and URA's official WhatsApp channel for a 60-day trial covering Domestic Taxes and Instant TIN support.
2. **Contact Centre Agent Assist Mode:** Deploy OmusoloSmart to Contact Centre agents as an internal co-pilot to instantly retrieve verified citations and calculate complex assessments during live calls.
3. **DRMS Regional Outreach Expansion:** Package the offline multimodal voice engine for deployment at mobile tax clinics, border posts, and regional tax education hubs across Uganda.

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